



Lesson 7.5 — Graphs of Sine and Cosine

Big idea: $y = \sin x$ and $y = \cos x$ are smooth waves that oscillate between -1 and 1 , with period 2π . Cosine is just sine shifted left by $\pi/2$.

Quick review

$y = \sin x$: starts at $(0, 0)$, peak at $(\pi/2, 1)$, back to 0 at $(\pi, 0)$, trough at $(3\pi/2, -1)$, back to 0 at $(2\pi, 0)$.

$y = \cos x$: starts at $(0, 1)$, down to 0 at $(\pi/2, 0)$, trough at $(\pi, -1)$, back to 0 at $(3\pi/2, 0)$, peak at $(2\pi, 1)$.

Both: period 2π , amplitude 1, range $[-1, 1]$, domain $\mathbb{R}$.

Symmetry: $\sin x$ odd (origin symmetry), $\cos x$ even (y-axis symmetry).

Phase shift relation: $\cos x = \sin(x + \pi/2)$.

Worked example

Worked example. List the x -values in $[0, 2\pi]$ where $\sin x = 1$, $\sin x = 0$, $\sin x = -1$.

$\sin x = 1$ at $x = \pi/2$. $\sin x = 0$ at $x = 0, \pi, 2\pi$. $\sin x = -1$ at $x = 3\pi/2$.

Warm-ups

1. Sketch (describe) $y = \sin x$ for one period from 0 to 2π . Where are the maximum and minimum?
2. Sketch $y = \cos x$ for one period. Where are the zeros?
3. What is the amplitude of $y = \sin x$?

Practice

1. List the x -values in $[0, 2\pi]$ where $\cos x = 0$.
2. Sketch (describe) $y = \sin x$ and $y = \cos x$ on the same axes from 0 to 2π . Identify two intersection points.
3. What is the difference (vertical) between $\sin x$ and $\cos x$ at $x = \pi/4$?

Challenge

1. Show graphically (in words) why $\cos x = \sin(x + \pi/2)$ — describe the shift.
2. Approximate $\sin(1.2)$ using the unit circle in radians (no calculator). $1.2 \text{ rad} \approx ?$



Answer key — Lesson 7.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. Max at $x = \pi/2$, min at $x = 3\pi/2$.
2. Zeros at $x = \pi/2$ and $3\pi/2$.
3. 1.

Practice

1. $x = \pi/2$ and $x = 3\pi/2$.
2. Two intersections in $[0, 2\pi]$: $x = \pi/4$ (both ≈ 0.707) and $x = 5\pi/4$ (both ≈ -0.707).
3. $\sin(\pi/4) = \sqrt{2}/2 = \cos(\pi/4)$. Difference 0.

Challenge

1. Cosine is sine shifted to the LEFT by $\pi/2$ (or equivalently, sine shifted RIGHT by $\pi/2$). So if you grab $\sin x$ and slide it left $\pi/2$, you get $\cos x$.
2. 1.2 rad is between $\pi/4 \approx 0.785$ and $\pi/3 \approx 1.047$ and $\pi/2 \approx 1.571$ — closest to $\pi/3 \approx 1.047$ with bit more.
 $\sin(\pi/3) = \sqrt{3}/2 \approx 0.866$. $\sin(\pi/2) = 1$. Linear interpolation between $\pi/3$ and $\pi/2$ at fraction $(1.2 - 1.047)/(1.571 - 1.047) \approx 0.292$. So $\sin(1.2) \approx 0.866 + 0.292(1 - 0.866) \approx 0.905$. (Calculator: 0.9320 — pretty close.)